ICE 10.2

Aman Gupta • Apr 10, 2025 (Edited Apr 27, 2025)

8 points

1. (2 marks) This is about breaking into the Diffie-Hellman key generation/exchange scheme. Let q=19, and alpha, a=3. In a particular instance Alice and Bob select private keys XA and XB, and based on that compute and exchange public keys YA=6, YB=7. Surely, since Alice and Bob know their respective private keys XA and XB, they can compute the shared key, K. BUT, can an intruder compute the shared key? In other words, given knowledge of [q=19, a=3, YA=6, YB=7] can an intruder figure out XA? If so, what is XA?

- 5
- 6
- 7
- **8 (correct)**
- 9
- None of the above

2. (2 marks) Returning to question 1, what is the shared key, K, computed by Alice or by Bob?

- 5
- 6
- 7
- 8
- 9
- 10
- **11 (correct)**
- None of the above

3. (2 marks) We propose to use ElGamal Cryptosystem for Bob to send messages to Alice, but using Diffie-Hellman generated shared key, K, using prime q = 11, and its primitive root a = 2. It the private keys XA and XB are such that shared key K = 5. Then what is the encrypted value Bob sends to Alice if the message M = 7?

- **2 (correct)**
- 3
- 4
- 5
- 6
- 7

4. (2 marks) Returning to question 3, above. What is the value of K^(-1) that Alice uses to decrypt

the received cipher and obtain M?

- 4
- 5

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- 8
- **9 (correct)**
- None of the above

ICE 10.2
Google Forms

Your work

Assigned

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Class comments

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